Experiment No. 5

BINARY TO GRAY AND GRAY TO BINARY CONVERTER

AIM:

To construct and verify the truth tables of

(i) Binary to gray code converter

(ii) Gray to binary code converter

COMPONENTS AND EQUIPMENTS REQUIRED:

Sl.No.	COMPONENT	SPECIFICATION	QTY.
1.	X-OR GATE	IC 7486	1
2.	IC TRAINER KIT	-	1
3.	Connecting wires	as required	

THEORY:

The availability of large variety of codes for the same discrete elements of information results in the use of different codes by different systems. A conversion circuit must be inserted between the two systems if each uses different codes for same information. Thus, code converter is a circuit that makes the two systems compatible even though each uses different binary code.

Here we take 4 bits for Gray to binary and vice versa conversions. Thus there are four inputs and four outputs. Gray code is a non-weighted code and its hamming distance is always 1.

The input variables for a Binary to Gray converter are designated as B3, B2, B1, and B0 and the output variables are designated as G3, G2, G1,and G0. From the truth table, combinational circuit is designed through K-Map.

Meanwhile the inputs for a Gray to Binary converter is G3, G2, G1,and G0 and its output generates B3, B2, B1, and B0.

DESIGN

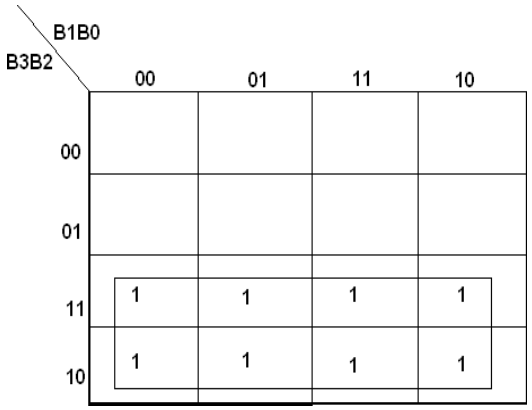
BINARY TO GRAY CODE CONVERTOR

TRUTH TABLE:

Binary inputs				**Gray outputs**			
B3	**B2**	**B1**	**B0**	**G3**	**G2**	**G1**	**G0**
0	**0**	**0**	**0**	**0**	**0**	**0**	**0**
0	**0**	**0**	**1**	**0**	**0**	**0**	**1**
0	**0**	**1**	**0**	**0**	**0**	**1**	**1**
0	**0**	**1**	**1**	**0**	**0**	**1**	**0**
0	**1**	**0**	**0**	**0**	**1**	**1**	**0**
0	**1**	**0**	**1**	**0**	**1**	**1**	**1**
0	**1**	**1**	**0**	**0**	**1**	**0**	**1**
0	**1**	**1**	**1**	**0**	**1**	**0**	**0**
1	**0**	**0**	**0**	**1**	**1**	**0**	**0**

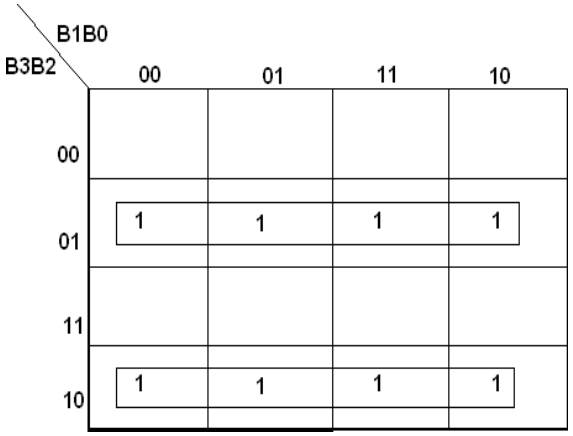
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

K-Map for G_3:



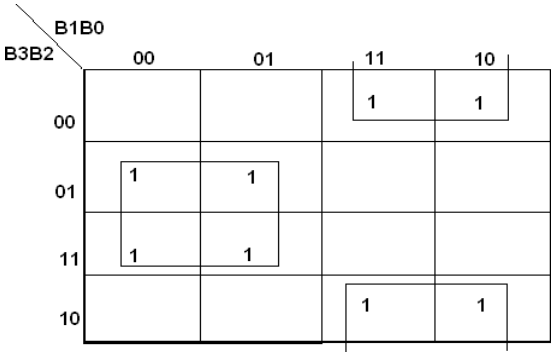
$G_3 = B_3$

K-Map for G_2:



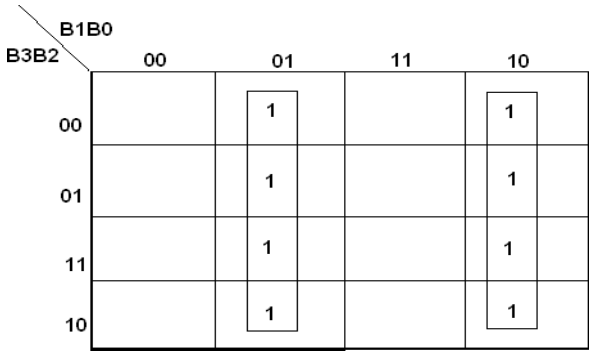
$G2 = \overline{B3}B2 + B3\overline{B2}$
$= B3 \oplus B2$

K-Map for G_1:



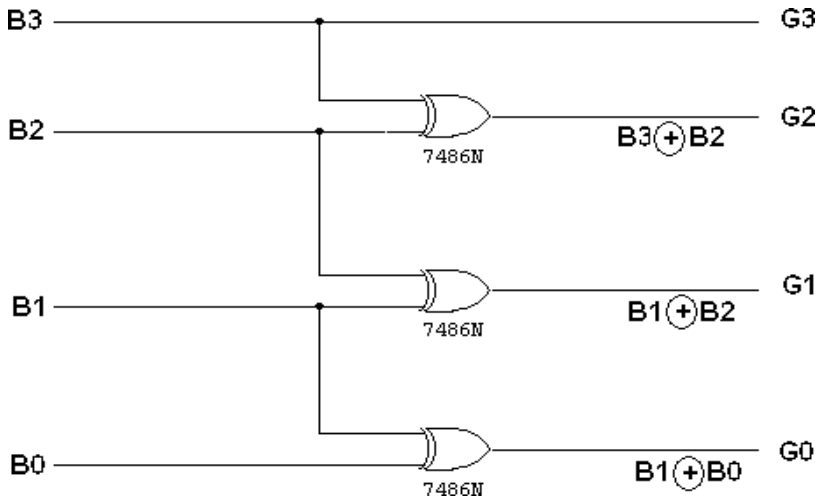
$G1 = \overline{B1}B2 + B1\overline{B2}$
$= B1 \oplus B2$

K-Map for G_0:



$$G0 = \overline{B1}B0 + B1\overline{B0}$$
$$= B1 \oplus B0$$

LOGIC DIAGRAM:-

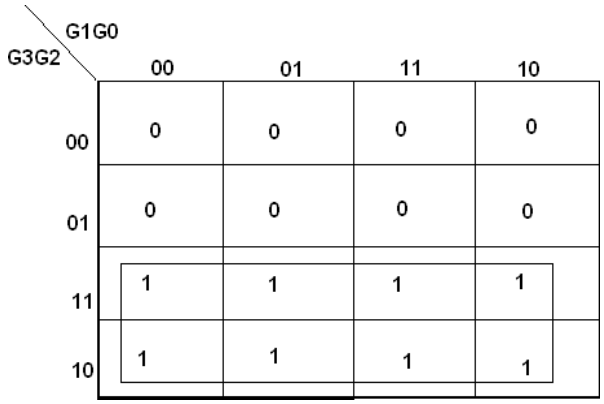


GRAY CODE TO BINARY CONVERTOR
TRUTH TABLE:

Gray Code				Binary Code			
G3	G2	G1	G0	B3	B2	B1	B0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	1	0	0	1	0
0	0	1	0	0	0	1	1
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
0	1	0	1	0	1	1	0
0	1	0	0	0	1	1	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	1	1	0	1	0
1	1	1	0	1	0	1	1
1	0	1	0	1	1	0	0
1	0	1	1	1	1	0	1

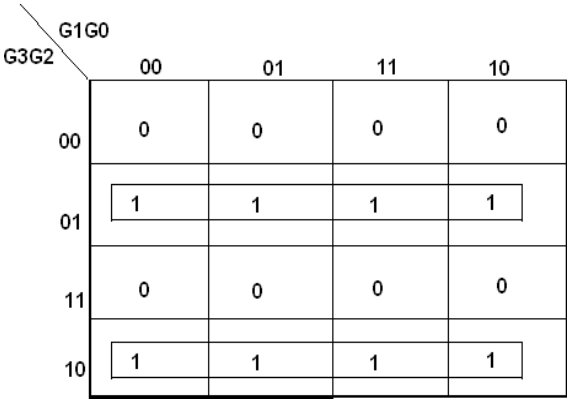
1 1	0 0	0 0	1 0	1 1	1 1	1 1	0 1

K-Map for B_3:



B3 = G3

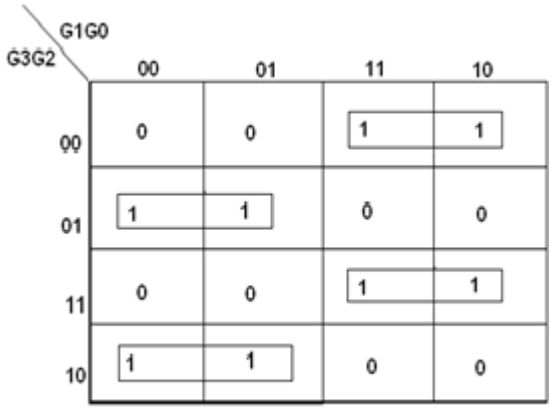
K-Map for B_2:



$B2 = \overline{G3}G2 + G3\overline{G2}$

$= G3 \oplus G2$

K-Map for B_1:



$B1 = \overline{G3}G2\overline{G1} + G3\overline{G2}\overline{G1} + \overline{G3}\overline{G2}G1 + G3G2G1$

$= \overline{G1}(\overline{G3}G2 + G3\overline{G2}) + G1(\overline{G3}\overline{G2} + G3G2)$

$= \overline{G1}.(G3 \oplus G2) + G1.\overline{(G3 \oplus G2)}$

$= G1 \oplus G2 \oplus G3$

$= G1 \oplus B2$

K-Map for B_0:

G3G2 \ G1G0	00	01	11	10
00	0	①	0	①
01	①	0	①	0
11	0	①	0	①
10	①	0	①	0

B0 = ----------

= ----------

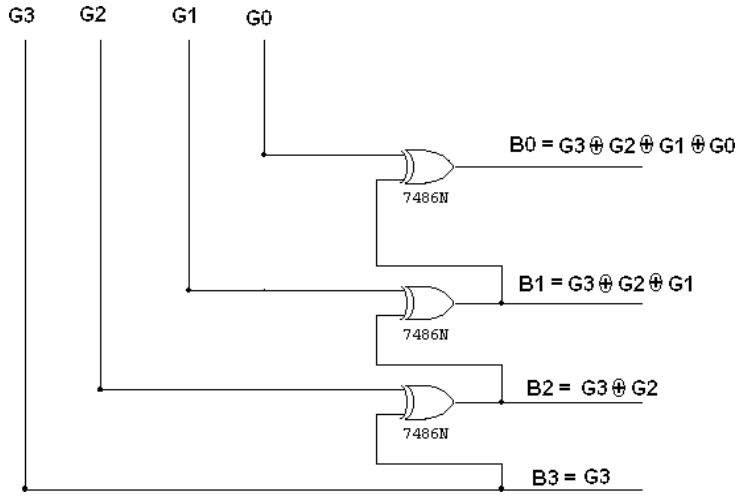
= ----------

= G3 ⊕ G2 ⊕ G1 ⊕ G0

= B1 ⊕ G0

(Solution should be completed by students)

LOGIC DIAGRAM OF GRAY TO BINARY CODE CONVERTER



PROCEDURE:

1. The circuit connections are made as shown in fig.
2. Pin (14) is connected to +Vcc and Pin (7) to ground.
3. In the case of binary to gray conversion, the inputs B0, B1, B2 and B3 are given at respective pins and outputs G0, G1, G2, G3 are taken for all the 16 combinations of the input.
4. In the case of gray to binary conversion, the inputs G0, G1, G2 and G3 are given at respective pins and outputs B0, B1, B2, and B3 are taken for all the 16 combinations of inputs.
5. The values of the outputs are tabulated.

RESULT:

Designed and implemented

a. Binary to gray code converter
b. Gray to binary code converter

Inference :

1. By using a single IC we can do both Gray to Binary and Binary to Gray converter.
2. Students get an idea to construct circuit for any code converter.